Module 3: Flower Species Principal Component Analysis Assignment

Summary Plots Discussion

Eigenvalue Table

Below is the Eigenvalue Table resulting from PCA, which analyzes patterns among various flower species based on petal width, petal length, sepal width, and sepal length. Row one shows Principal Component 1, which explains the majority of variance among the flower species. The cumulative percentage for this component alone is 72.77%. The eigenvalue is the largest of any eigenvalue (2.91), indicating that PC1 explains more variation in the dataset than any other principal component. PC2 accounts for 23.03% of the variance, and when considered in conjunction with PC1, the cumulative percentage is 95.80%. PC1, PC2, and PC3 capture nearly all the variance in the dataset (99.49%). Row four's eigenvalue is close to zero, which shows it captures almost no variance in the data. Because they account for over 95% of the variance, PC1 and PC2 are the most important components in this analysis.

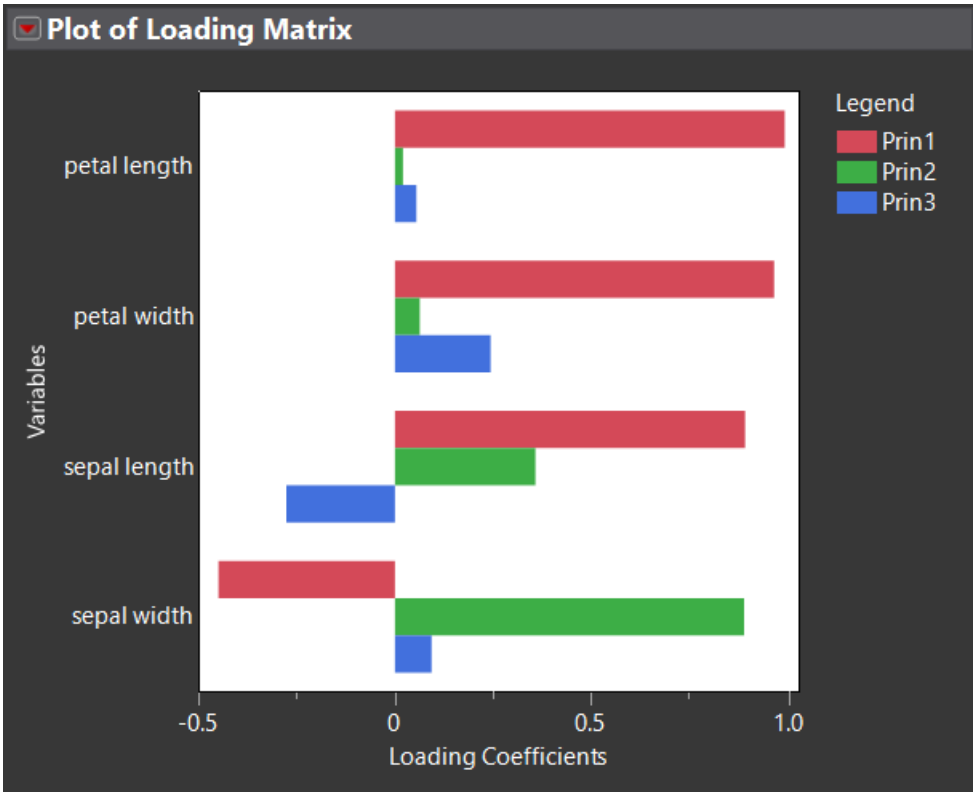
Eigenvalues

Number	Eigenvalue	Percent	20 40 60 80	Cum Percent
1	2.910818	72.770		72.770
2	0.921221	23.031		95.801
3	0.147353	3.684		99.485
4	0.020608	0.515		100.000

Loading Plot

The loading plot shows PC1 in red, PC2 in green, and PC3 in blue. The variables are shown on the y-axis, and the x-axis represents the loading coefficients, which show how much each variable contributes to the principal component. PC1 is most influenced by petal length, petal width, and sepal length, and has a slight negative influence from sepal width. PC2 variability, however, is most influenced by sepal width, followed by sepal length. This plot shows that PC1 captures variation in petal dimension, while sepal length has mixed

contributions to variance in both PC1 and PC2, and sepal width is the primary variable contributing to PC2.



Loading Matrix

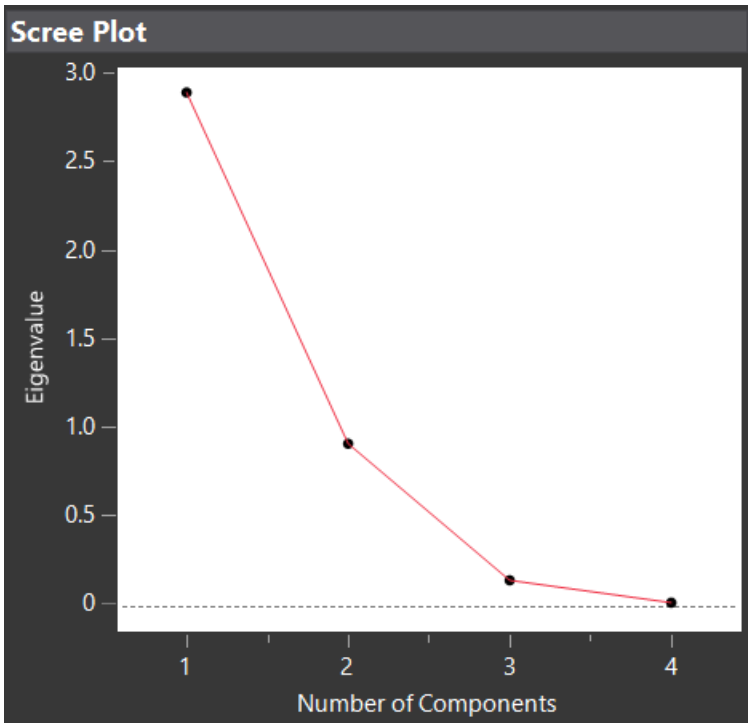
The loading plot above is quantified in the loading matrix below, where the rows represent the variables and each column corresponds to a principal component. Coefficients with an absolute value close to one contribute more to the component, and if positive, it has a positive relationship with the principal component. A negative loading coefficient indicates a negative relationship with the principal component. Values near zero do not influence the component. PC1 is shown to be influenced by petal length (0.99), petal width (0.965), and sepal length (0.89). Sepal width is negatively correlated to PC1, but the relationship is weak (-0.44). PC2 is primarily defined by sepal width (0.888), followed by sepal length (0.357).

Loading Matrix

	Prin1	Prin2	Prin3	Prin4
sepal length	0.89122	0.35735	-0.27677	-0.03761
sepal width	-0.44931	0.88835	0.09291	0.01782
petal length	0.99168	0.02025	0.05408	0.11501
petal width	0.96500	0.06279	0.24329	-0.07516

Scree Plot

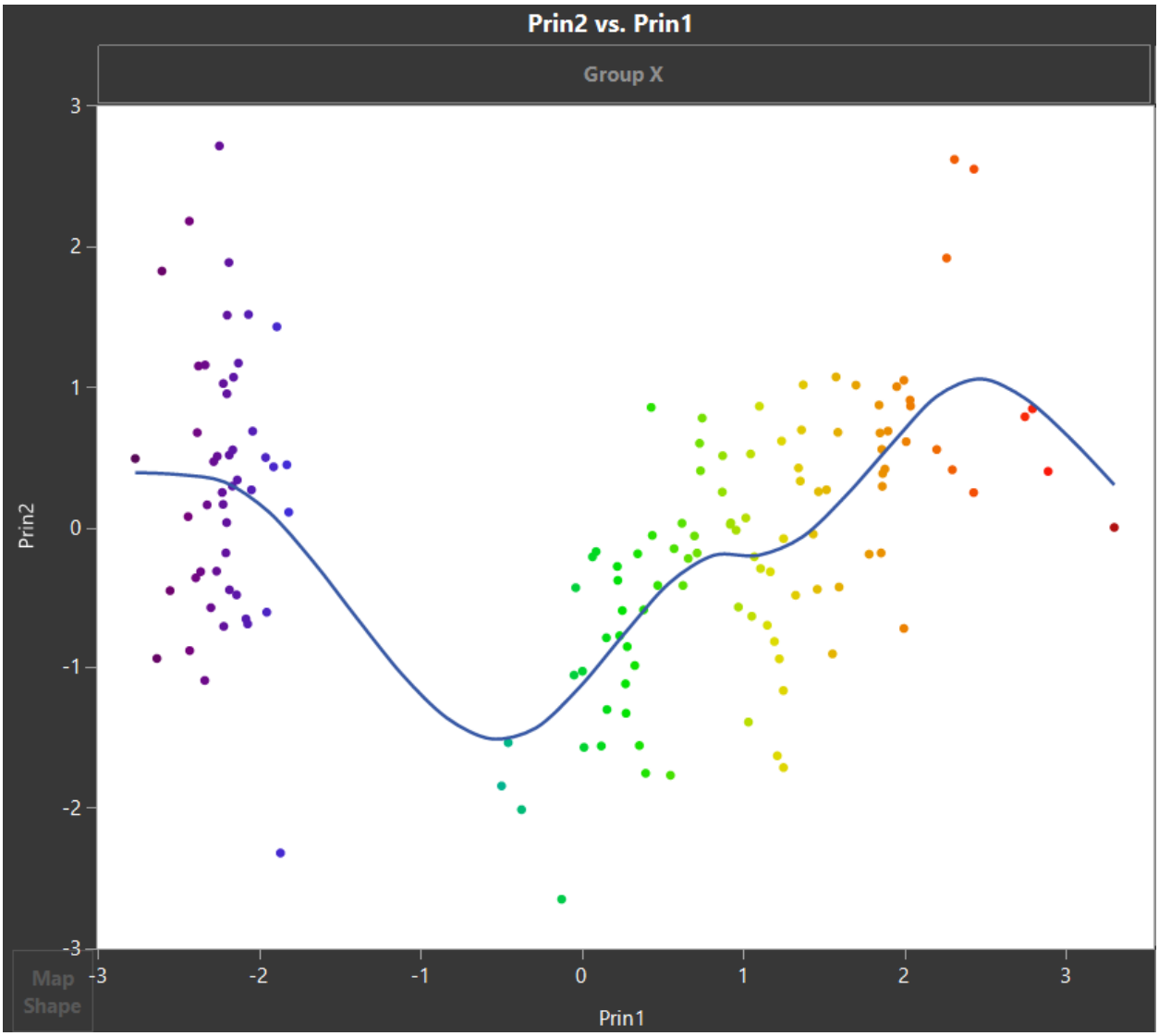
The scree plot below shows the principal components on the x-axis and eigenvalues on the y-axis. A larger eigenvalue indicates the component explains more variance in the data. The plot shows the eigenvalues decreasing as the number of components increases: PC1 explains most of the variance in flower dimensions, followed by PC2, PC3 and PC4. The eigenvalues for PC3 and PC4 are so close to zero, that these components contribute almost no meaningful variance. PC1 and PC2 capture most of the variance in the data.



Principal Components Scatterplot Discussion

PCA has shown that PC1 and PC2 together explain 95.8% of the variance in flower dimensions. Below, they are graphed together, with PC1 on the x-axis and PC2 on the y-axis. The scatterplot allows us to identify patterns and clusters among flower samples, indicating groups of flower species. There is a clear separation along the PC1 axis between the cluster on the left (represented by purple and dark blue points) and the group on the right (green, yellow, orange, and red points), indicating that there are at least two distinct species in the dataset. The species on the left is likely characterized by smaller petal length and width, but variation in sepal length and width. The cluster on the right has a visual gradient that suggests a species of flower that has wide dimensional variation, or two species that are closely related, with overlap in appearance. The line's curve also has two

humps, which support the assertion that there are two clusters within the group on the right side, one represented by the green points and the other by the yellow and orange points. The data appear to represent three different species of flowers, based on the observed clusters.



Conclusion

PCA of the flower dataset revealed that PC1 accounts for the majority (72.77%) of the variance in flower dimensions. The variation of this component is heavily influenced by petal length, width, and sepal length. PC2 accounts for an additional 23.03% of variance, and is primarily contributed to by sepal width. Cumulatively, PC1 and PC2 account for 95.80% of the variance in flower dimension, but the third and fourth components contribute very little variation. This shows how most of the variation among the four-dimensional measurements can be represented in two by PC1 and PC2. This simplification allows for better visualization of patterns and clusters, which distinguishes three species within the data.